

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

College of Computing and Data Science

SC2006 – Software Engineering

Lab 2 Deliverables

SCSJ Group 3

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Use Case Diagrams

Note: Diagrams are grouped by subsystem first, complete Use Case Diagram below

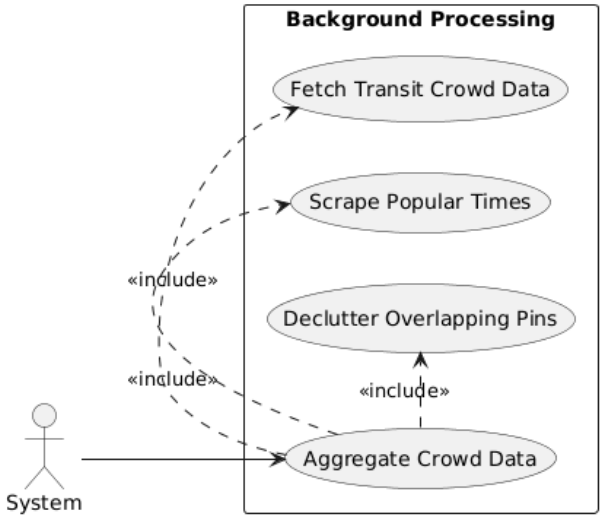


Fig 1. Background Processing (System)

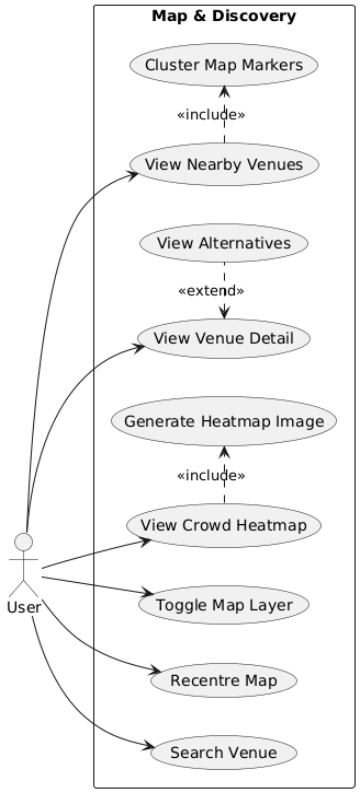


Fig 2. Map & Discovery (User)

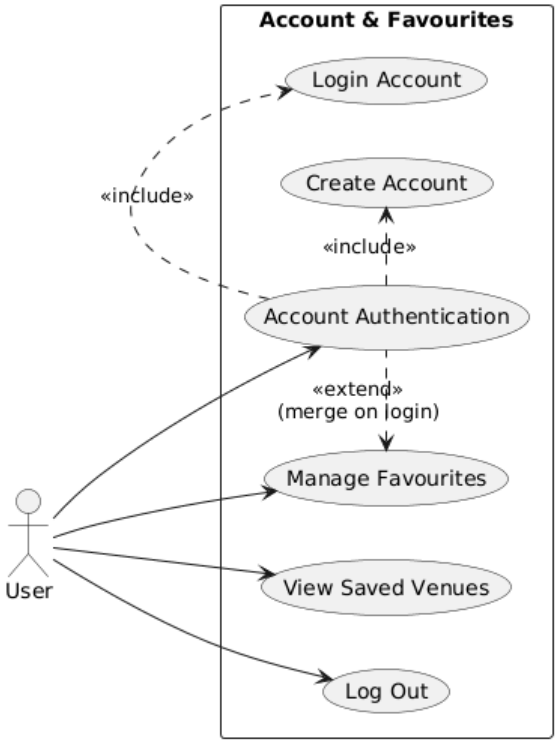


Fig 3. Account & Favourites (User)

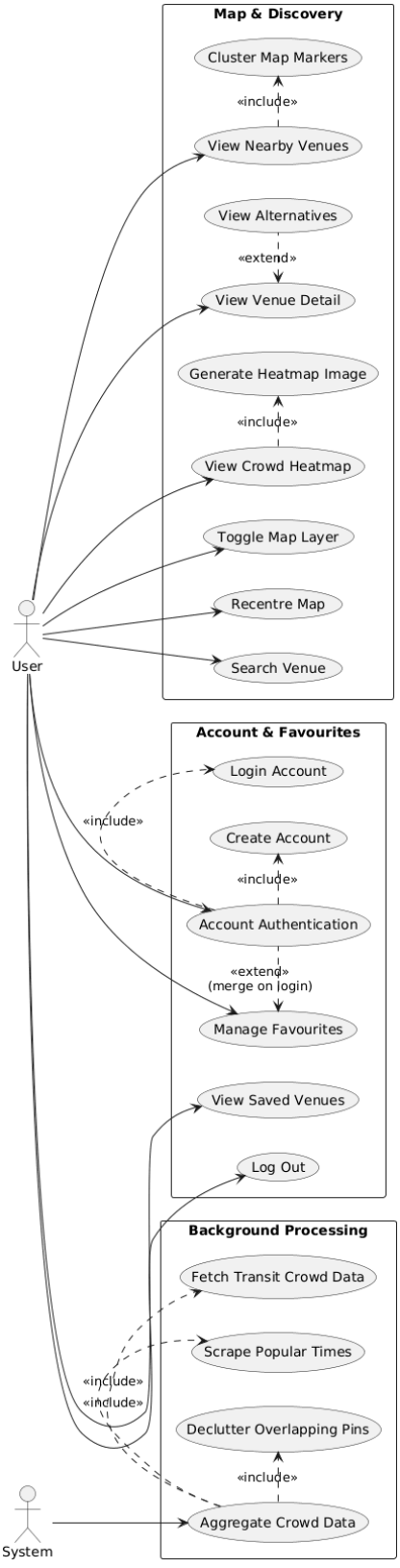


Fig 4. Complete Use Case Diagram

Use Case Descriptions

For Functional Requirement #1 – Background crowd data aggregation

Use Case ID:	#1-1		
Use Case Name:	AggregateCrowdData		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	System
Description	Periodically collects crowd data from external sources and merges it into one unified venue list.
Preconditions	None (runs on a schedule / cache expiry).
Postconditions	An up-to-date, merged venue list is available to serve users.
Priority	High
Frequency of Use	High (every 1–5 minutes for transit; ~daily for mall data)
Flow of Events	1. System checks whether cached transit data has expired (5 min). 2. If expired, invokes FetchTransitCrowdData. 3. System checks whether cached Popular Times data is stale (~daily). 4. If stale, invokes ScrapePopularTimes per venue. 5. System merges transit and mall data into one list. 6. System invokes DeclutterOverlappingPins before serving it.
Alternative Flows	If a source fails to respond, the system continues serving its last successfully cached result for that source.
Exceptions	If a source fails for a subset of records, only that subset is skipped.
Includes	FetchTransitCrowdData, ScrapePopularTimes, DeclutterOverlappingPins
Special Requirements	Requires a valid LTA DataMall API account key.
Assumptions	None
Notes/Issues	None

Use Case ID:	#1-2		
Use Case Name:	FetchTransitCrowdData		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	System
Description	Fetches real-time MRT/LRT crowd density from LTA DataMall.
Preconditions	Cached transit data has expired.
Postconditions	Current crowd levels for all tracked stations are retrieved.
Priority	High
Frequency of Use	High (every 5 minutes)
Flow of Events	1. System calls LTA DataMall's Station Crowd Density API. 2. System maps each station's raw crowd code to a Low/Moderate/High level. 3. System checks each station's real operating hours; a closed station is marked Closed regardless of the raw reading.
Alternative Flows	None
Exceptions	If the API is unreachable, previously cached data continues to be served.
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	LTA's feed has no notion of "closed" — derived separately from real per-station first/last train times.

Use Case ID:	#1-3		
Use Case Name:	ScrapePopularTimes		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	System
Description	Retrieves live "Popular Times" crowd data for a mall/venue, since no official API exists for it.
Preconditions	Cached Popular Times data for the venue is stale, or an on-demand refresh was requested.
Postconditions	The venue's current crowd % and weekly pattern are updated.
Priority	High
Frequency of Use	Daily (batch), plus on-demand per venue
Flow of Events	1. System drives a headless browser to search for the venue on Google Maps. 2. System opens the venue's place page. 3. System scrolls to load the Popular Times chart. 4. System extracts the full week's hourly pattern and opening hours. 5. System caches the result.
Alternative Flows	None
Exceptions	If the chart never loads, the venue falls back to its last cached reading.
Includes	None
Special Requirements	Requires a headless browser (Puppeteer) runtime on the server.
Assumptions	None
Notes/Issues	Inherently fragile, depends on an unofficial page layout Google can change without notice.

Use Case ID:	#1-4		
Use Case Name:	DeclutterOverlappingPins		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	System
Description	Repositions venues sharing a near-identical map location so each remains individually visible and selectable.
Preconditions	A merged list of venues with coordinates is available.
Postconditions	No two venues occupy the same small map area.
Priority	Medium
Frequency of Use	High
Flow of Events	1. System groups venues by rounding coordinates to a small grid cell. 2. For each group with more than one venue, system computes the shared centre point. 3. System arranges the group's venues in a small circle around that centre.
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Distinct from ClusterMapMarkers, which is a zoom-dependent display grouping, not a coordinate fix.

For Functional Requirement #2 – View nearby venues

Use Case ID:	#2-1		
Use Case Name:	ViewNearbyVenues		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Displays a map and list of nearby venues (transit stations and malls) with current crowd levels.
Preconditions	User has granted location permission, or the system falls back to a default location.
Postconditions	Map and list show venues near the user, each with a crowd level.
Priority	High
Frequency of Use	High
Flow of Events	1. App opens. 2. System requests current location. 3. System retrieves nearby venues via AggregateCrowdData. 4. System displays venues as colour-coded pins on the map. 5. System displays the same venues as a distance-sorted list below the map. 6. As the user pans/zooms, the list updates to match the visible area.
Alternative Flows	If location permission is denied, the map centres on a default location (Marina Bay).
Exceptions	If no venues are available, an empty-state message is shown.
Includes	AggregateCrowdData, ClusterMapMarkers
Special Requirements	Device location services (GPS).
Assumptions	None
Notes/Issues	Both layers (Transit/Venue) are off by default — nothing is drawn until the user picks one.

Use Case ID:	#2-2		
Use Case Name:	ClusterMapMarkers		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User (included by ViewNearbyVenues)
Description	Groups nearby map markers into a single clustered pin at low zoom levels to reduce clutter.
Preconditions	Map is displaying two or more venues close together at the current zoom.
Postconditions	Overlapping markers appear as one clustered pin with a count.
Priority	Medium
Frequency of Use	High
Flow of Events	1. System evaluates on-screen marker positions. 2. Markers that would visually overlap are grouped into one clustered pin. 3. If the user zooms in far enough, the cluster expands into individual pins. 4. If the user taps a cluster, the map zooms into that area.
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Distinct from DeclutterOverlappingPins (repositions venues sharing near-identical real coordinates, at the data layer).

Use Case ID:	#2-3		
Use Case Name:	ToggleMapLayer		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Shows or hides Transit or Venue pins on the map.
Preconditions	None
Postconditions	The selected layer's pins appear or disappear on the map and matching list.
Priority	Medium
Frequency of Use	High
Flow of Events	1. User taps a layer toggle button. 2. System shows/hides that category's pins and list rows.
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Both layers are off by default.

Use Case ID:	#2-4		
Use Case Name:	RecentreMap		
Created By:	Joseph Tan	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Recentres the map on the user's current location.
Preconditions	User has moved the map away from their location.
Postconditions	Map recentres on the user's current GPS location.
Priority	Low
Frequency of Use	Medium
Flow of Events	1. User taps the recentre button. 2. System retrieves current GPS location. 3. Map animates back to centre on it.
Alternative Flows	None
Exceptions	If location is unavailable, an alert is shown.
Includes	None
Special Requirements	Device location services (GPS).
Assumptions	None
Notes/Issues	None

For Functional Requirement #3 – View venue detail

Use Case ID:	#3-1		
Use Case Name:	ViewVenueDetail		
Created By:	Xu Hanwen	**Last Updated By:**	Xu Hanwen
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Displays full crowd detail for a selected venue.
Preconditions	User has selected a venue from the map, list, or search results.
Postconditions	Venue's full detail (crowd %, weekly histogram, opening hours, source) is displayed.
Priority	High
Frequency of Use	High
Flow of Events	1. User selects a venue. 2. System retrieves the venue's full detail. 3. System displays name, category, address, crowd %, crowd level, weekly histogram, and opening hours. 4. If a lower-crowd alternative exists nearby, system invokes ViewAlternatives.
Alternative Flows	None
Exceptions	If the venue's data is no longer available, a "not found" message is shown.
Includes	None (extended by ViewAlternatives)
Special Requirements	None
Assumptions	None
Notes/Issues	A "Check crowd now" action forces a fresh Popular Times scrape for this one venue, bypassing the cache.

Use Case ID:	#3-2		
Use Case Name:	ViewAlternatives		
Created By:	Xu Hanwen	**Last Updated By:**	Xu Hanwen
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Recommends lower-crowd venues of the same category near the currently viewed venue. Extends ViewVenueDetail.
Preconditions	User is viewing a venue's detail screen.
Postconditions	Lower-crowd alternatives, if any, are displayed.
Priority	Medium
Frequency of Use	Medium
Flow of Events	1. System identifies venues of the same category. 2. System filters to those within 3km. 3. System filters to those with a lower crowd percentage. 4. System sorts remaining venues by distance. 5. System displays the results.
Alternative Flows	If the user selects an alternative, ViewVenueDetail is invoked for it.
Exceptions	If no qualifying alternative exists, nothing is shown in this section.
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Transit venues are excluded from alternatives matching.

For Functional Requirement #4 – Crowd heatmap

Use Case ID:	#4-1		
Use Case Name:	ViewCrowdHeatmap		
Created By:	Mao Yikai	**Last Updated By:**	Mao Yikai
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Displays a blurred, colour-graded overlay of crowd levels across all tracked venues.
Preconditions	User is viewing the map.
Postconditions	The heatmap overlay is shown or hidden on the map.
Priority	Medium
Frequency of Use	Medium
Flow of Events	1. User taps the heatmap toggle. 2. System invokes GenerateHeatmapImage if no fresh image is cached. 3. System places the image on the map, positioned to its real geographic bounds. 4. User taps the toggle again to hide it.
Alternative Flows	None
Exceptions	If image generation fails, the toggle has no visible effect.
Includes	GenerateHeatmapImage
Special Requirements	iOS only, react-native-maps has no gradient/heatmap layer outside Google Maps, which Expo Go no longer supports.
Assumptions	None
Notes/Issues	Closed or not-yet-scraped venues contribute no colour.

Use Case ID:	#4-2		
Use Case Name:	GenerateHeatmapImage		
Created By:	Mao Yikai	**Last Updated By:**	Mao Yikai
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	System
Description	Renders the crowd heatmap as a single blurred, colour-graded image.
Preconditions	ViewCrowdHeatmap was invoked and no fresh cached image exists.
Postconditions	A PNG image representing current crowd levels across all open venues is available.
Priority	Medium
Frequency of Use	On-demand, cached ~5 minutes
Flow of Events	1. System filters to open venues with real crowd data. 2. System projects each venue's coordinates onto a canvas. 3. System draws a soft radial colour gradient per venue over a neutral base wash. 4. System exports the canvas as a PNG.
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	Rendered via the same headless browser used for ScrapePopularTimes.
Assumptions	None
Notes/Issues	Closed/unavailable venues are excluded so they don't falsely appear "quiet."

For Functional Requirement #5 – Search

Use Case ID:	#5-1		
Use Case Name:	SearchVenue		
Created By:	K Eswaran	**Last Updated By:**	K Eswaran
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Lets the user search the full tracked venue list by name.
Preconditions	None
Postconditions	Matching venues are displayed, best match first.
Priority	High
Frequency of Use	High
Flow of Events	1. User types a venue name. 2. System filters the full known venue list by name match. 3. Results starting with the query are ranked above ones that merely contain it. 4. User selects a result, invoking ViewVenueDetail.
Alternative Flows	None
Exceptions	If no venues match, a "no results" message is shown.
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Purely client-side against the already-loaded venue list, no separate search API call.

For Functional Requirement #6 – Accounts & favourites

Use Case ID:	#6-1		
Use Case Name:	AccountAuthentication		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Teo Kok Wei, Oscar
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Let the user sign up for a new account or log into an existing one.
Preconditions	None
Postconditions	User is authenticated and holds a session token.
Priority	Medium
Frequency of Use	Low
Flow of Events	1. User opens Settings and selects Sign Up or Log In. 2. User enters a username and password. 3. System validates credentials. 4. If the user had locally-saved favourites, ManageFavourites is invoked to offer merging them. 5. System returns the user to Settings, now signed in.
Alternative Flows	None
Exceptions	Wrong username/password shows the same generic error either way, so login attempts can't be used to check which usernames exist.
Includes	LoginAccount, CreateAccount
Special Requirements	Passwords hashed with bcrypt. Sessions are JWT bearer tokens (30-day expiry).
Assumptions	None
Notes/Issues	Username-based (not email). A deliberate simplification from Lab 1's original plan.

Use Case ID:	#6-2		
Use Case Name:	LoginAccount		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Teo Kok Wei, Oscar
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Allows a user to log in to an existing account.
Preconditions	User has an existing account.
Postconditions	User is authenticated and returned to Settings.
Priority	High
Frequency of Use	Low
Flow of Events	1. User enters username and password. 2. System validates credentials against the database. 3. If correct, system issues a session token.
Alternative Flows	None
Exceptions	If credentials are incorrect, a generic error is shown.
Includes	None
Special Requirements	None
Assumptions	User has already registered.
Notes/Issues	None

Use Case ID:	#6-3		
Use Case Name:	CreateAccount		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Allows a visitor to create a new account.
Preconditions	None
Postconditions	A new account is created and stored in the database.
Priority	High
Frequency of Use	Low
Flow of Events	1. User enters a username and password. 2. System validates the username format (3–24 letters/numbers/underscore) and password length (8+ characters). 3. System checks the username isn't already taken. 4. System hashes the password and stores the new account. 5. System issues a session token.
Alternative Flows	None
Exceptions	If the username is already taken, or the input is invalid, an error is shown.
Includes	None
Special Requirements	Password stored only as a bcrypt hash, never in plain text.
Assumptions	None
Notes/Issues	None

Use Case ID:	#6-4		
Use Case Name:	ManageFavourites		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Save or remove venues as favourites, and view them in a dedicated list.
Preconditions	None
Postconditions	The venue is added to or removed from favourites.
Priority	Medium
Frequency of Use	Medium
Flow of Events	1. User taps the favourite icon on a venue. 2. If signed in, system saves/removes the favourite against the account. 3. If not signed in, system saves/removes it in local on-device storage. 4. If the user later signs in with local favourites present, they are explicitly prompted (not silently merged) to bring them into the account.
Alternative Flows	None
Exceptions	If a favourited venue no longer exists in the data source, it is silently dropped from the list.
Includes	None (extended by AccountAuthentication)
Special Requirements	None
Assumptions	None
Notes/Issues	The explicit merge prompt exists to prevent bleeding of one guest's leftover local saves into every subsequent account signed into on the same device.

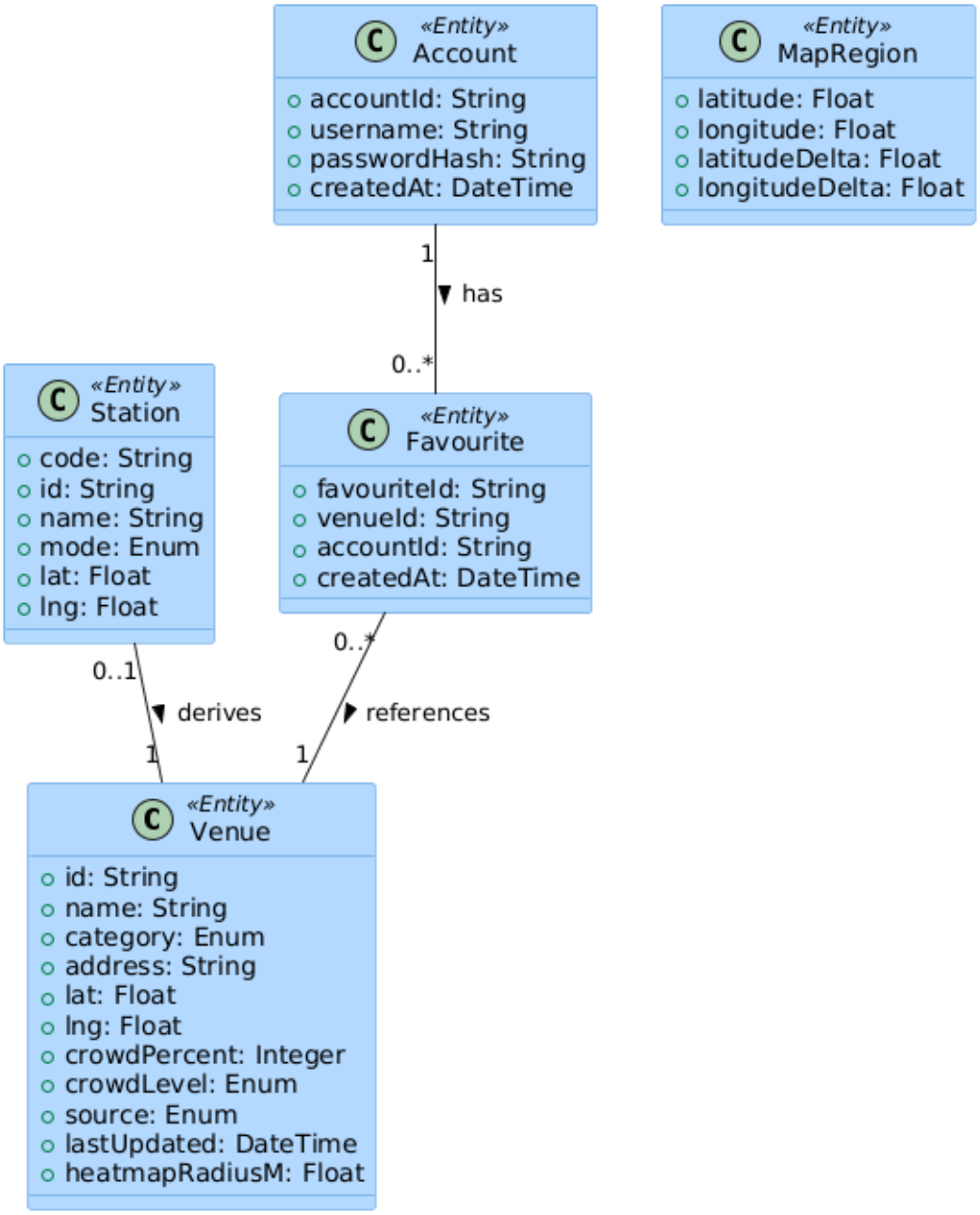
Use Case ID:	#6-5		
Use Case Name:	ViewSavedVenues		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Joseph Tan
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

Actor	User
Description	Displays all venues the user has favourited.
Preconditions	None
Postconditions	A list of saved venues, each with its current crowd level, is displayed.
Priority	Medium
Frequency of Use	Medium
Flow of Events	1. User selects the Saved tab. 2. System retrieves the user's favourited venue ids (account or local, depending on sign-in state). 3. System displays them sorted by distance.
Alternative Flows	None
Exceptions	If no venues are saved, an empty-state message is shown.
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	None

Use Case ID:	#6-6		
Use Case Name:	LogOut		
Created By:	Teo Kok Wei, Oscar	**Last Updated By:**	Teo Kok Wei, Oscar
Date Created:	16/09/2026	**Date Last Updated:**	16/09/2026

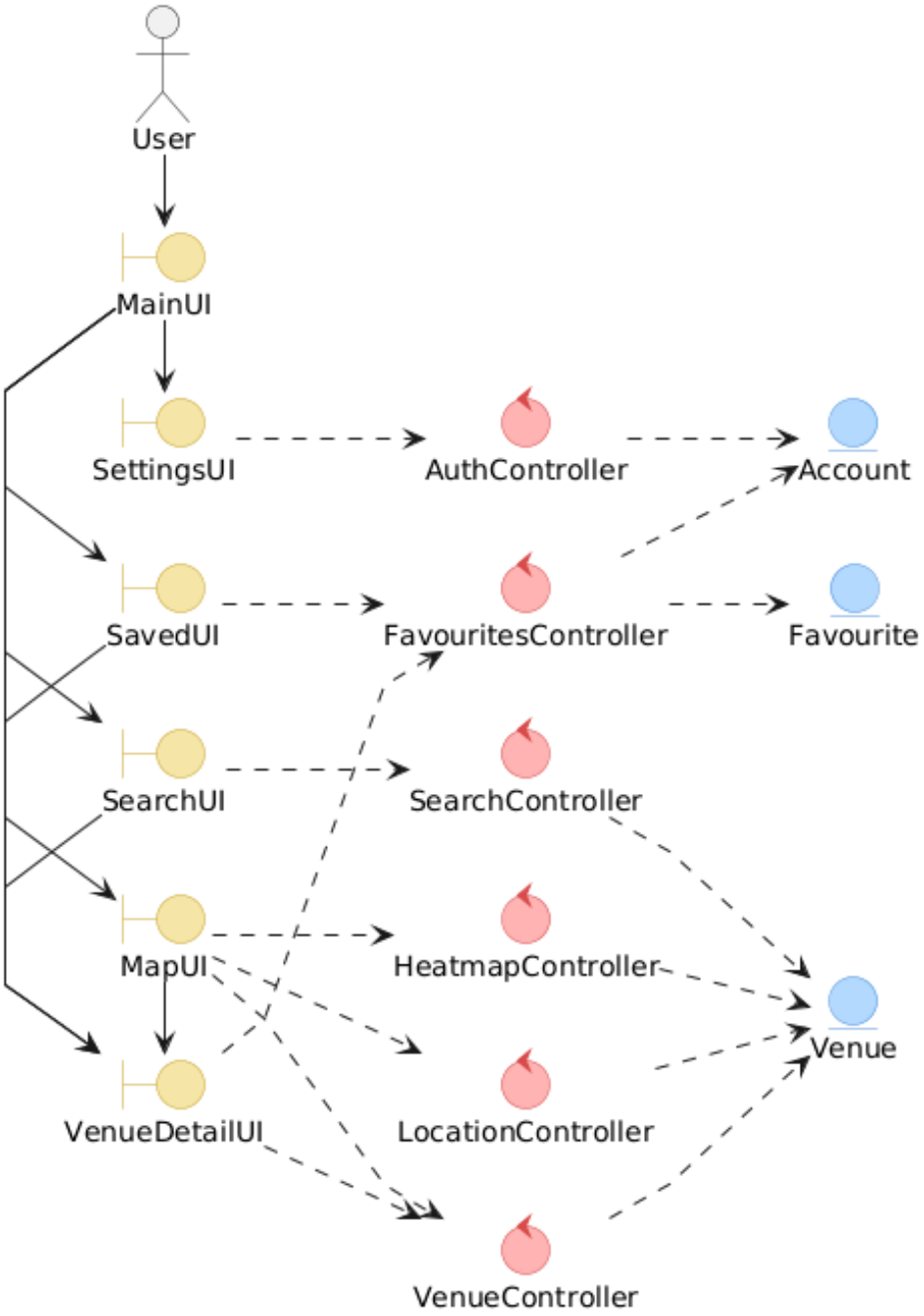
Actor	User
Description	Ends the current session.
Preconditions	User is signed in.
Postconditions	Session token is discarded; user reverts to guest, on-device-only state.
Priority	Low
Frequency of Use	Low
Flow of Events	1. User selects Log Out in Settings. 2. System discards the stored session token. 3. Saved venues now read from local storage instead of the account.
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes/Issues	Also triggered automatically if the backend ever rejects the session token as expired/invalid.

Class Diagram of Entity Classes



Note: MapRegion is a transient, unpersisted value object, hence no relationship lines

Key Boundary Classes and Control Classes



Initial Dialog Map

